Project Documentation – Employee Laptop Request

Project	ServiceNow-Based Employee Laptop Request Application
Platform	ServiceNow – custom application with a Service Portal front end
Application	HR Onboarding Application (Update Set)
Custom table	Laptop Request Form (u_laptop_request_form)
Record Producer	Laptop Request
Date	22 October 2026

1. Introduction

- **Project Title:** ServiceNow-Based Employee Laptop Request Application
- **Team Members:** Pasapu Vineela – Developer (individual project)

2. Project Overview

- **Purpose:** Employees often ask for laptops through emails and informal messages, so requests get lost, are repeated, and nobody can see their status. This application provides one Service Portal form where an employee submits a laptop request. The system enforces the one-request-per-year policy, routes the request to the IT Manager for approval and shows the status to the end.
- **Objectives:**
  - Make laptop requests quick, with no need to open backend forms.
  - Capture the same key details every time: employee, request type, reason and status.
  - Store every request in one custom table with a unique number.
  - Enforce one laptop request per employee per year.
  - Route requests for approval and track status as Pending Approval, Approved or Rejected.
  - Limit access by role: IT Manager (view-only) and IT Employee (create and manage).
- **Features:**
  - Custom application and table with auto-numbered records.
  - Record Producer “Laptop Request” that creates records in the custom table.
  - Request types: Basic (personal or general office use) and Advance (software development or technical work).
  - Business rule that blocks a second request by the same employee in the same year.
  - Approval flow in Flow Designer, with the status updated automatically.
  - Roles, groups and ACLs for secure access.
  - Deployment through an Update Set exported as XML.

Request lifecycle: Pending Approval → Approved or Pending Approval → Rejected

3. Architecture

End-to-end flow:

Employee → Service Portal → Record Producer → Laptop Request table → Business Rule → Flow Designer approval → Status tracking

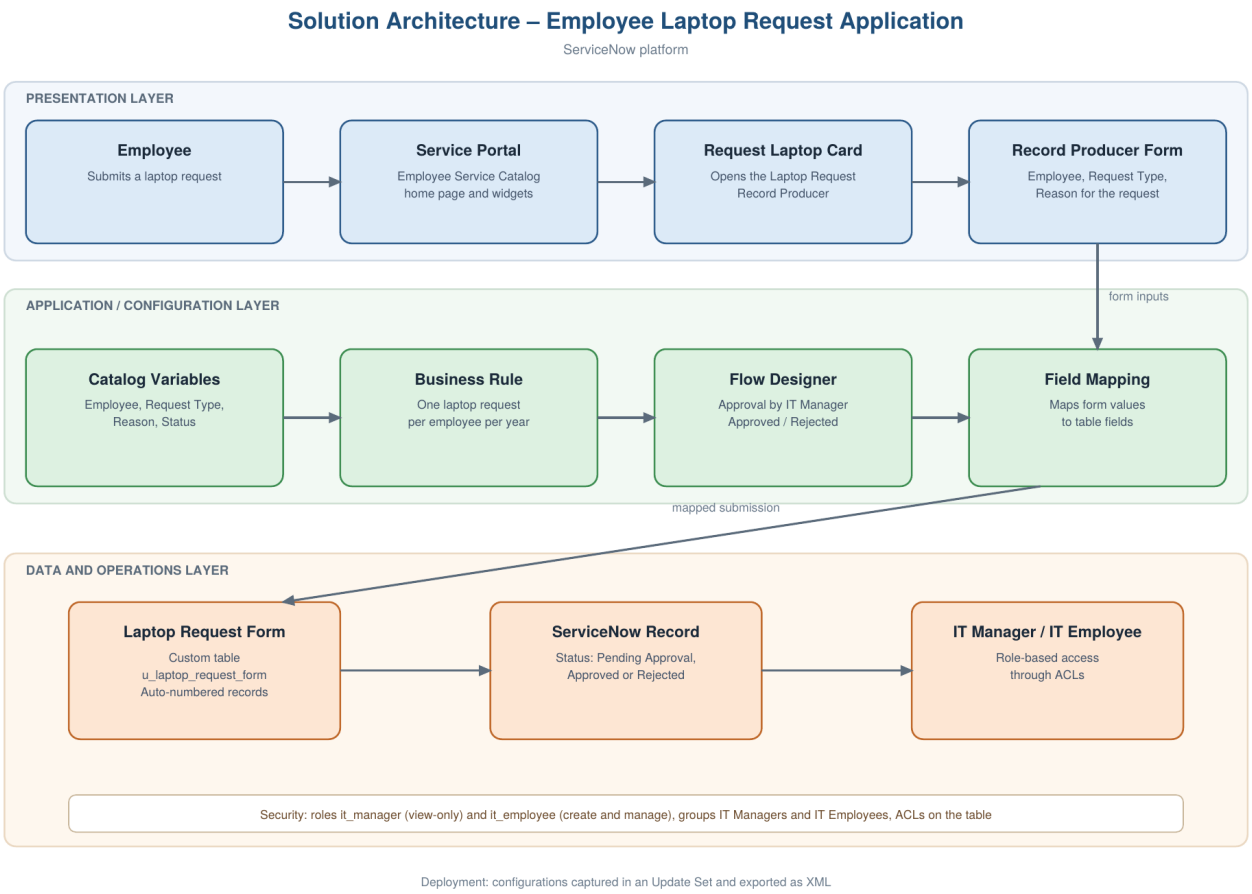


Figure 1: Solution architecture of the Employee Laptop Request Application

- **Frontend:** Service Portal. The home page shows the “Our Laptop Services” panel and the “Request a Laptop” card. Request Laptop opens the Laptop Request form through the Service Catalog.
- **Backend:** Custom application built in Studio. The Record Producer writes into the custom table, a business rule validates the one-request-per-year policy, and a Flow Designer flow handles approval. Roles and ACLs handle security.
- **Database:** The table “Laptop Request Form” holds one record per request. Its columns are listed below.

Column	Type	Status	Purpose
Number	String (autonumber)	To implement	Unique request ID, default getNextObjNumberPadded()
Employee	Reference → User	To implement	Employee who raised the request
Request Type	Choice	To implement	Basic or Advance
Reason	String	To implement	Justification for the request
Status	Choice	To implement	Pending Approval, Approved or Rejected
Approver	Reference → User	Recommended	IT Manager who decides on the request
Created / Updated / Updated by	Date-Time, String	System	Audit information kept by the platform

4. Setup Instructions

- **Prerequisites:**
  - A ServiceNow instance (for example a Personal Developer Instance) with the admin role.
  - Access to Studio, Service Catalog, Service Portal and Flow Designer.

- Test users, plus IT Manager and IT Employee groups for testing.

- **Installation:** Step-by-step guide to build the application.

1. Create an Update Set named “HR Onboarding Application” and set it as Current before starting development.

2. Open ServiceNow Studio and create the custom application “Project Application”, defining its scope and access.

3. Create the table “Laptop Request Form” with the columns listed in Section 3 and turn on auto-numbering.

4. Create the Record Producer “Laptop Request” with the table as its target. Add the variables (employee, request type, reason) and map them to table columns. Create the catalog item name “Employee Service Catalog” and category “Service”.

5. Create the business rule that blocks a second request in the same year.

6. Build the approval flow in Flow Designer so that the IT Manager approves or rejects and the status is updated.

7. Create the roles it_manager and it_employee, the groups IT Managers and IT Employees, and the ACLs.

8. Add the Record Producer to the portal page and test the complete flow.

5. Folder Structure

A ServiceNow application has no folder tree. Its components are grouped inside the application, as shown below.

Area	Components
Application (Studio)	Project Application, the container for everything below
Data	Table “Laptop Request Form” with columns, choice lists and dictionary entries
Service Catalog	Record Producer “Laptop Request”: variables and field mapping
Service Portal	Portal record, home page, request page and widgets
Server-side and security	Business rule, Flow Designer approval flow, roles, groups and ACLs
Deployment	Update Set “HR Onboarding Application” exported as XML

6. Running the Application

- **Employee:** Open the Service Portal and log in. Select Request Laptop on the “Request a Laptop” card. Complete the Laptop Request form (request type and reason) and select Submit. The new record gets the next request number and the status Pending Approval.
- **IT Manager:** Open the approval request, review the reason and approve or reject it. The status updates automatically. The manager has view-only access to the records.
- **IT Employee:** Open the Laptop Request Form list to create and manage requests.
- **Administrator:** In the Record Producer, select Try It to preview the form without opening the portal.

7. API Documentation

The project adds no custom REST API. Requests are created through the Record Producer. Records can also be read or written through the standard ServiceNow Table API, subject to ACLs. The table name is u_laptop_request_form.

Method	Endpoint	Description
GET	/api/now/table/u_laptop_request_form	List requests (supports sysparm_query and sysparm_limit)
GET	/api/now/table/u_laptop_request_form/{sys_id}	Retrieve one request
POST	/api/now/table/u_laptop_request_form	Create a request
PATCH	/api/now/table/u_laptop_request_form/{sys_id}	Update fields such as status

8. Authentication

- **Authentication:** Employees sign in to the Service Portal with their ServiceNow accounts (local, or SSO/LDAP if the instance uses it). The platform manages the session, and the application adds no separate token handling.
- **Authorization:** Roles and ACLs decide who can read, create and update records.
  - IT Employees (it_employee): create and manage laptop requests.
  - IT Managers (it_manager): view-only access to requests, and approve or reject through the approval flow.
  - Administrators: manage the table, Record Producer, portal and configuration.
- **API access:** REST calls use basic authentication or OAuth, and the same ACLs apply.

9. User Interface

- **Portal home page:** The “Our Laptop Services” panel sits beside the “Request a Laptop” card, which holds the Request Laptop button.

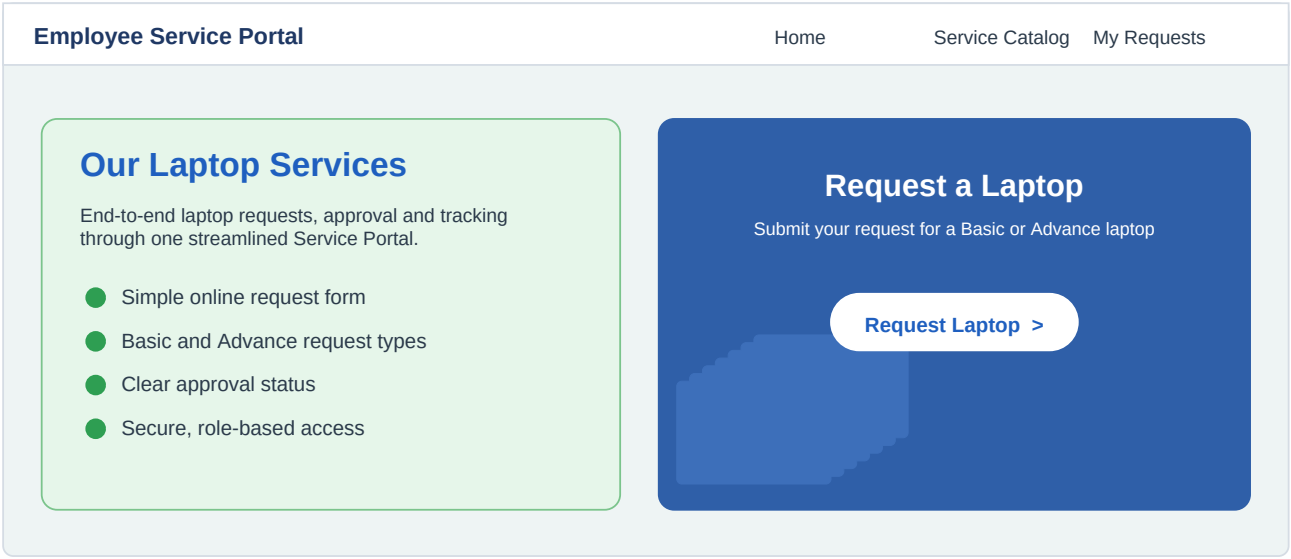


Figure 2: Service Portal home page

- **Laptop Request form:** The Record Producer form shows these fields, with Save as Draft and Submit buttons on the right.
  - Employee (user lookup)
  - Request Type: Basic, Advance
  - Reason for the request (text)
  - Status: Pending Approval (set automatically, read-only)

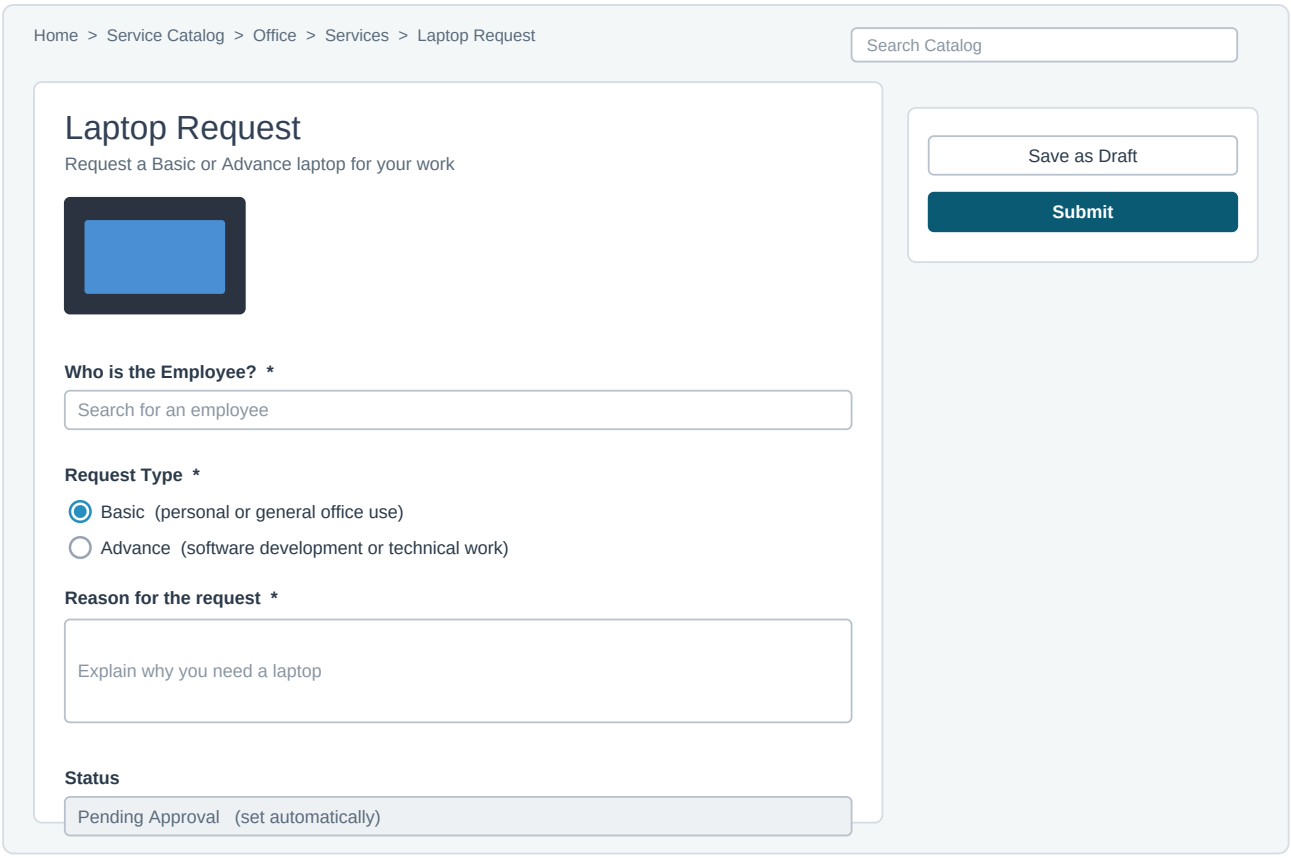


Figure 3: Laptop Request form on the portal

10. Testing

Testing covers the employee's view, the IT Manager's view and the IT Employee's view. Tools to use: the portal itself, impersonation of test users, the list and form views, and the system logs. Record the result of each case in the Status column.

No.	Test case	Expected result	Status
1	Submit a valid laptop request through the portal	A record is created in Laptop Request Form with status Pending Approval	To verify
2	Leave a required field empty	Form blocks submission and shows a message	To verify
3	Select Basic and then Advance	The request type is saved correctly on the record	To verify
4	Compare variables with record fields	Each variable maps to the correct column	To verify
5	Submit a second request as the same employee in the same year	Business rule blocks it and shows a message	To verify
6	IT Manager approves a request	Status changes to Approved	To verify
7	IT Manager rejects a request	Status changes to Rejected	To verify
8	Log in as IT Manager, IT Employee and employee	Each sees only what roles and ACLs allow	To verify

11. Screenshots or Demo

Figures 2 and 3 (Section 9) show the employee-facing portal. The figures below show the back-end configuration. All figures are illustrative mock-ups of the planned configuration; replace them with screenshots from your instance when the build is complete.

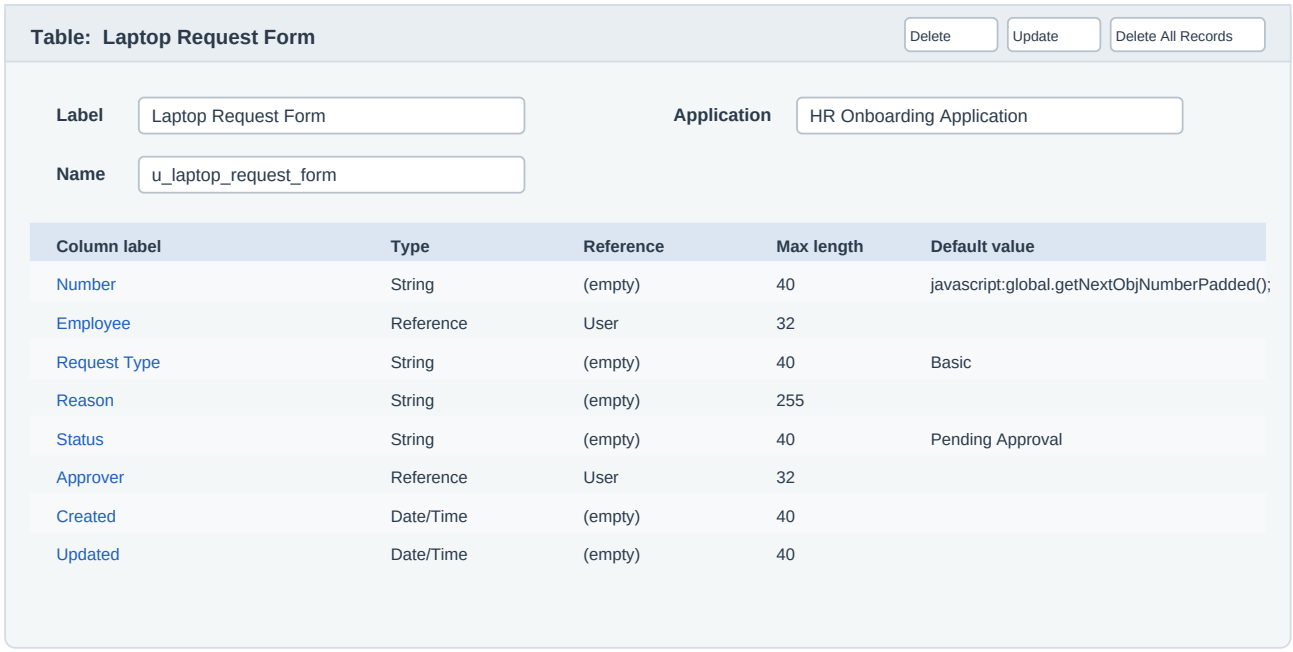


Figure 4: Custom table "Laptop Request Form" and its columns

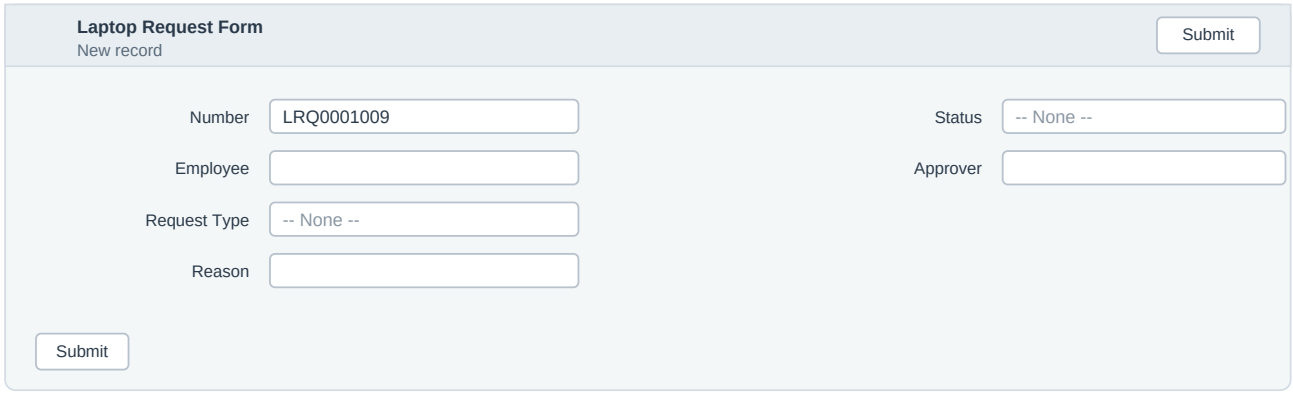


Figure 5: New record form with auto-generated number (LRQ0001009)

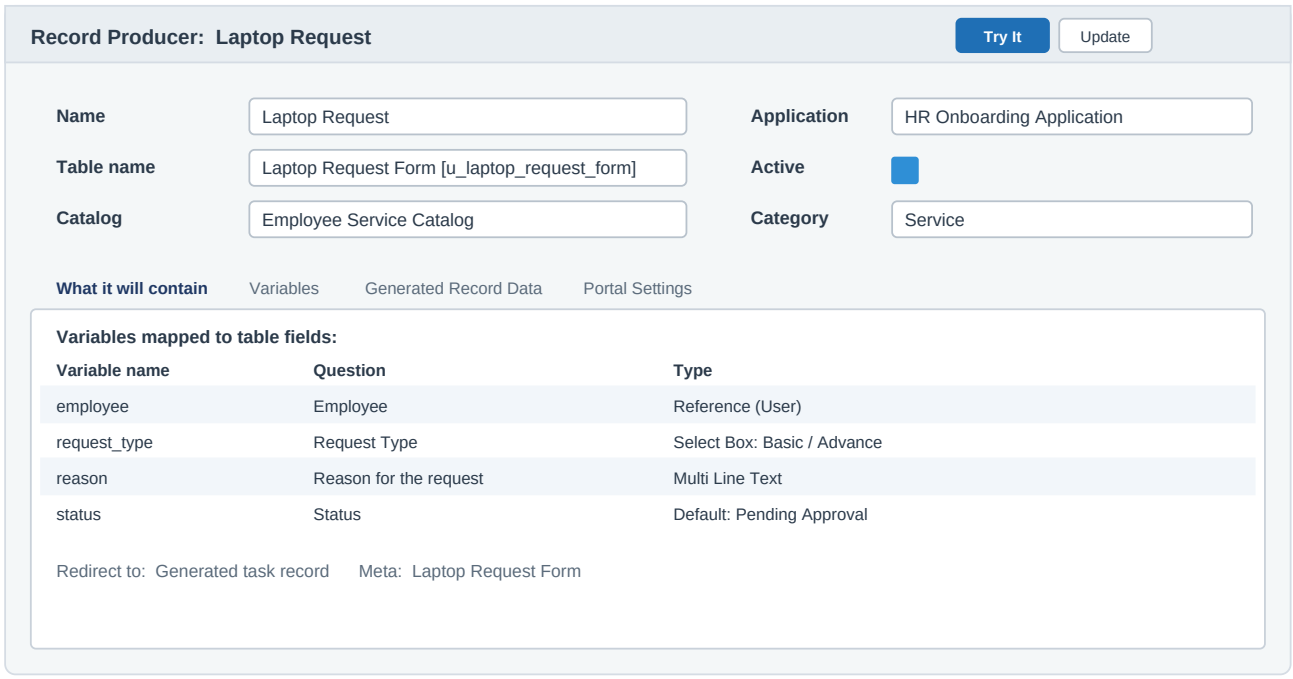


Figure 6: Record Producer "Laptop Request" linked to the custom table

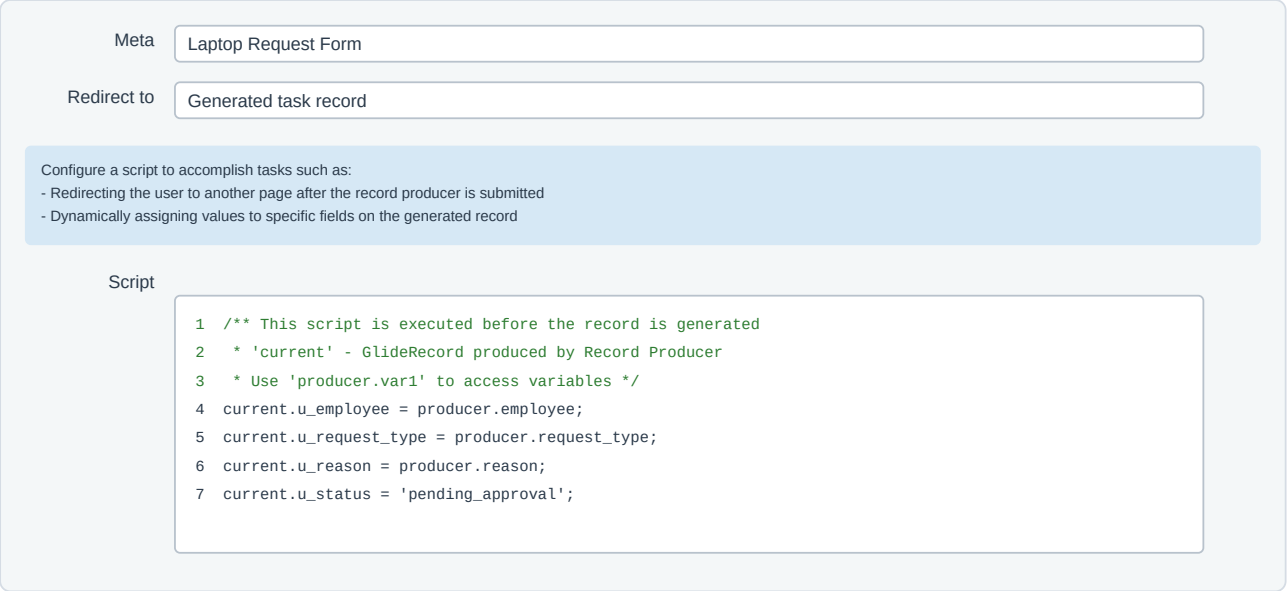


Figure 7: Record Producer redirect setting and script

12. Known Issues

Points worth reviewing in the configuration:

- **Status on the form:** Requesters must not be able to set Status. Keep it read-only and default it to Pending Approval.
- **Choice fields:** Request Type and Status should be Choice fields so the database enforces valid values.
- **Annual check:** The business rule counts records created in the current calendar year. Decide whether rejected requests should count towards the limit.
- **Naming:** Some write-ups still use the names “HR Onboarding Application” and “Project Application”. Confirm the application and update set names you want to present.
- **Placeholder text:** Replace the default Record Producer short description with real guidance for the employee.

13. Future Enhancements

- Email and SMS notifications for submission, approval and rejection.
- SLA timers and automatic escalation for requests waiting on approval.
- A “My Requests” portal page where employees track status and add comments.
- Integration with asset management for laptop procurement and tracking.
- Dashboards for request volume, request type trends and approval time.
- Knowledge-base suggestions while the employee types a reason.
- A mobile-friendly view or the Now Mobile app.